# OPTIMIZATION OF LAST-MILE DELIVERY USING BIG DATA AND SAP ERP SYSTEM

*The project stands out by combining Python-based data-driven insights with SAP ABAP simulation, using real-world datasets, which is rarely explored in academic research.*

**By**

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## Introduction

The current world of rapid retailing requires businesses to have smart systems to stay in pace with the evolving demand and delivery requirements. In this type of project, SAP ABAP allows simulating a real-life logistics scenario, such as returns, peak days, and demand spikes, using real retail data. The simulation improves delivery time, cost efficiency, and vehicle usage with simulation within a real ERP environment by transforming unprocessed data into intelligent decisions. It is the place where technology meets logistics to plan smarter supply chain.

## Research Question

How can real-time big data analytics integrate with data-driven demand forecasting to improve last-mile delivery efficiency in urban retail supply chains using SAP in emerging economies?

## Objective

1. To assess how real-time big data analytics and data-driven demand forecasting are being used (or underused) in urban last-mile delivery systems within emerging markets.
2. To categorize and identify the main drawbacks of the current last-mile delivery simulation models, particularly their inability to integrate real-time input, focus on KPIs, and adjust to uncertainty in the real world.
3. To develop a simulation model using SAP system that integrates real-time traffic and demand data to optimise logistics KPIs, including delivery time, cost, and vehicle usage.
4. To evaluate the performance of the SAP simulation under varied conditions, such as peak vs. off peak hours and varying levels of data quality (high vs. low).
5. To provide strategic recommendations for logistics decision-makers in emerging economies on implementing integrated analytics and forecasting through customised ERP models.

## Methodology

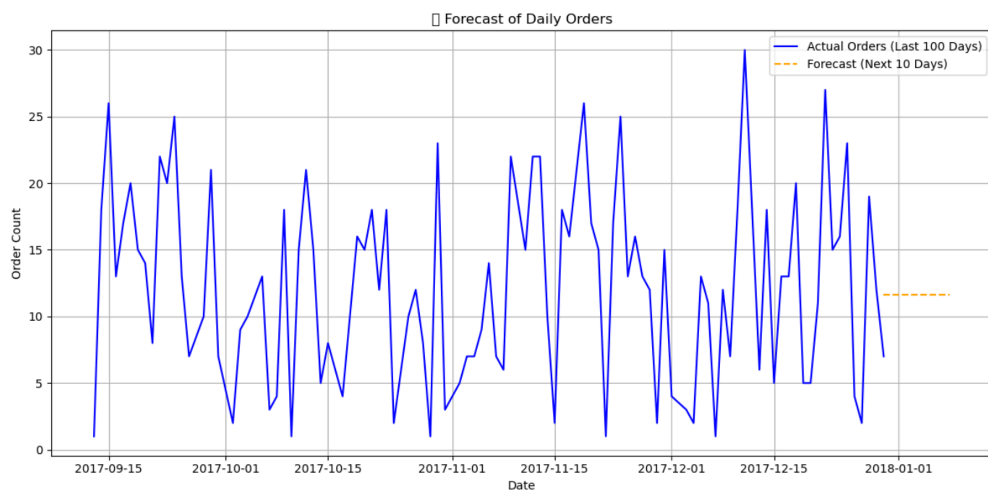
01 A total of 96 articles were reviewed using a Systematic Literature Review (SLR) approach using PRISMA to explore current research in logistics simulation and SAP technologies.

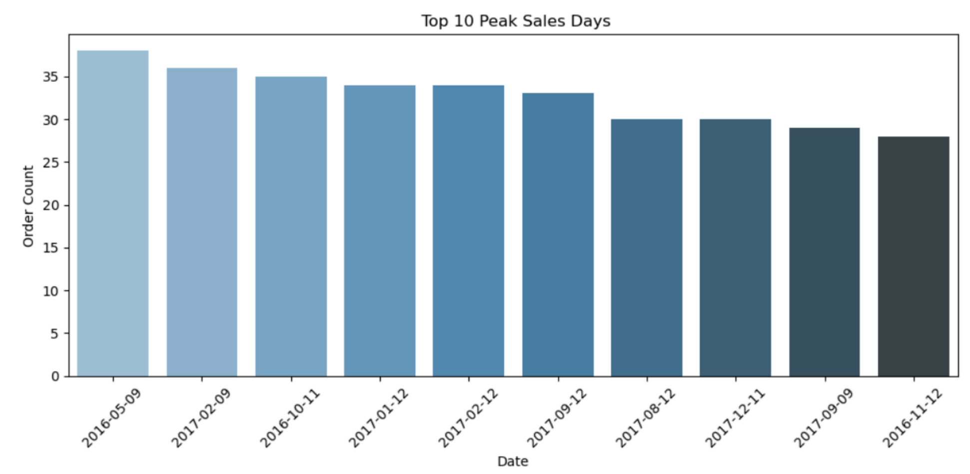
02 Retail logistics data was analysed using Python to extract important details such as demand trends, returns, and delivery delays.

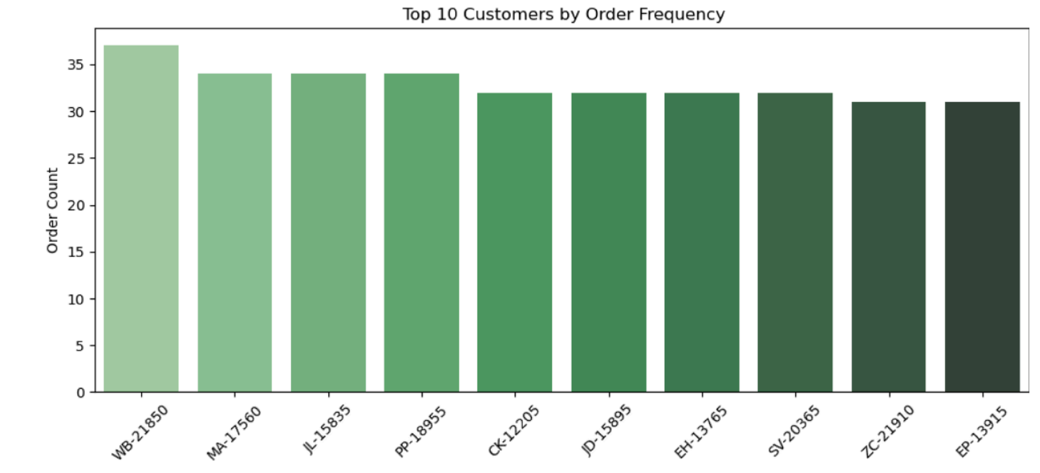
03 The extracted insights were used to build a simulation model in SAP ABAP that calculated delivery time, cost, and vehicle usage

04 Real-world factors like peak demand days and product returns were added to make the simulation model more realistic and accurate.

05 The simulation results were used to evaluate how input data quality affects logistics performance and decision-making outcomes.

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ABAP Simulation on Retail Dataset forecast

| Date | Forecast | Time | Cost | Vehicles |
|---|---|---|---|---|
| 31-12-2017 | 12.06 | 39.12 | 66.33 | 0 |
| 01-01-2018 | 12.06 | 24.12 | 60.30 | 0 |
| 02-01-2018 | 12.06 | 24.12 | 60.30 | 0 |
| 03-01-2018 | 12.06 | 39.12 | 66.33 | 0 |
| 04-01-2018 | 12.06 | 24.12 | 60.30 | 0 |
| 05-01-2018 | 12.06 | 24.12 | 60.30 | 0 |
| 06-01-2018 | 12.06 | 24.12 | 60.30 | 0 |
| 07-01-2018 | 12.06 | 39.12 | 66.33 | 0 |
| 08-01-2018 | 12.06 | 39.12 | 66.33 | 0 |
| 09-01-2018 | 12.06 | 24.12 | 60.30 | 0 |

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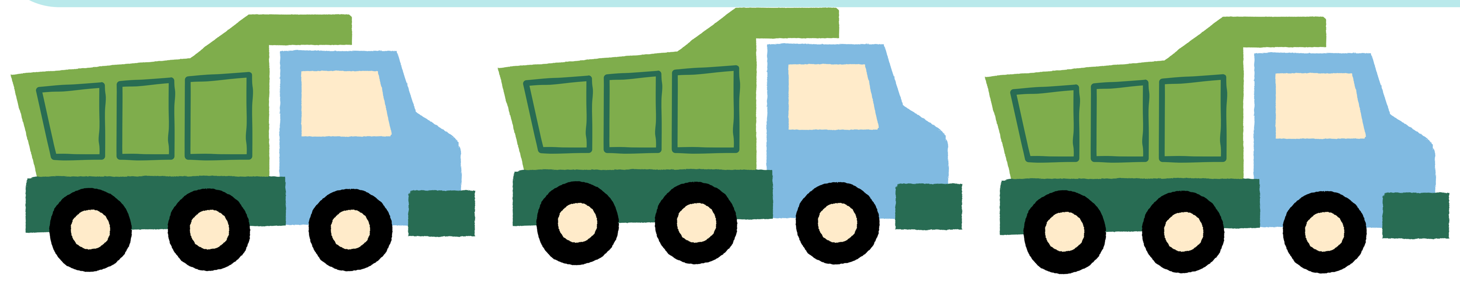
## Results

## Gap Identified

1. The majority of previous research concentrate on either delivery or data, but rarely both at once.
2. While SAP's simulation tools are often used in industrial scenario planning and teaching, logistics planning has never made use of them.
3. There are no simulation systems that evaluate the effects of real-time data and forecasts on delivery KPIs, including delivery time and cost.



## Limitations

1. Limited use of SAP System
2. Limited Resources Availability on this topic
3. Basic KPI Focused
4. Use of freely available dataset.
5. Lack of real-time data integration such as API, GPS
6. Lack sfeedback from logistics managers/stakeholder on the simulation result

## Conclusion

This study demonstrated how combining real-time data with simulation models can enhance last-mile delivery planning in new markets. Through data preprocessing in Python and simulation with SAP ABAP, it was demonstrated that logistics KPIs such as delivery time, cost, and vehicle usage can be optimised. The results revealed the weaknesses of the current models and demonstrated that even simple tools, when applied appropriately, could provide realistic and scalable logistics data. Overall, the project offers a convenient and flexible approach to data-driven logistics simulation on SAP.

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